Context-Aware Control for Mecanum-Wheeled Robots in Intelligent Service Environments

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***Abstract*—This paper presents a unified context-aware navigation framework for Mecanum-wheeled service robots operating in indoor warehouse and service environments. The system integrates four tightly coupled layers: (i) a perception layer fusing LiDAR, depth-camera, and YOLO v8 data; (ii) a context detection module (CASG) that classifies three operational contexts—Open Zone (OZ), Narrow Corridor (NC), and Human-Proximate Zone (HPZ); (iii) an A* global planner with semantic cost augmentation; and (iv) a Context-Aware Nonlinear Model Predictive Controller (CA-NMPC) whose cost-function weights and safety constraints adapt in real time to the detected context. A comparative study of four global planners (A*, Hybrid A*, Theta*, and RRT*) confirms A* as the best choice for indoor warehouse navigation. Simulation in Gazebo Fortress yields 100 % task-completion rate and, compared with an A*+DWA baseline, reduces average trajectory jerk by 34.2 % and improves minimum human clearance by 41.8 %. Real-world experiments on a physical 4-wheel Mecanum robot running ROS2 Humble achieve a 94.7 % success rate across 38 runs with zero collisions.**

***Index Terms*—Mecanum wheel robot, context-aware control, A*, CA-NMPC, nonlinear MPC, NMPC, service robotics, ROS2**

I. Introduction

Service robots deployed in hospitals, airports, and smart warehouses must navigate safely among humans, narrow passages, and dynamically changing layouts. Holonomic Mecanum-wheeled platforms are particularly well-suited to these environments: they can translate in any direction and rotate in place without chassis reconfiguration, greatly reducing dead-lock situations in confined spaces [1].

However, most current autonomous navigation stacks—including ROS2 Nav2 [2]—treat planning and control as decoupled, *static-parameter* pipelines. A global planner (typically A* or Dijkstra) computes an obstacle-free reference path from a fixed occupancy grid; a separate local planner (DWA or TEB) then tracks this path using uniform obstacle-inflation radii and a single velocity window. The CA-NMPC controller then adjusts velocity commands with no awareness of the current operating context. Two key failure modes arise in service environments:

1) **Context insensitivity**: appropriate velocity limits and safety radii change as the robot transitions between an open lobby, a narrow corridor, and a human-occupied zone—but static planners cannot adapt without manual reconfiguration.
2) **Semantic unawareness**: standard costmaps treat a human and a shelf post identically, yielding socially inappropriate and potentially unsafe trajectories.

This paper addresses both limitations through a framework that propagates a *semantic context signal* from the perception layer directly into the CA-NMPC controller, enabling the robot to adapt its trajectory optimisation to the operating environment in real time.

Paper Contributions

1) A **Context Detection module** (CASG) that fuses LiDAR geometry, depth-camera range, and real-time YOLO v8 detection to classify three operational contexts (OZ, NC, HPZ) with <10 ms latency.
2) A **CA-NMPC controller** with a context-parameterised stage-cost matrix $Q(c)$ and context-dependent safety constraints as the *sole* trajectory controller, solved at 25–30 Hz on embedded hardware.
3) A **comparative study** of four global planners (A*, Hybrid A*, Theta*, RRT*) and three controllers (PID, LMPC, CA-NMPC).
4) **Full ROS2 Humble implementation** validated in simulation and on a physical Mecanum robot (38 runs).

II. Related Work

A. Global Path Planning

A* [3] explores a 2-D occupancy grid by expanding nodes that minimise $f(n) = g(n) + h(n)$, where $g(n)$ is the cost-to-come from start and $h(n)$ is an admissible heuristic. With the Euclidean heuristic the algorithm guarantees optimality on unit-cost grids. Zakir et al. [4] compare A*, Dijkstra, and RRT in 2-D warehouse maps, finding A* paths of 9.37 m versus RRT paths of 11.63 m with planning times of 372 ms versus 641 ms. Mi et al. [5] combine A* with Jump Point Search (JPS) to reduce node traversal by 37.6 % while preserving optimality.

*Hybrid A** [6] augments the standard grid search with a continuous-state heading dimension, generating paths that respect the kinematic constraints of car-like or differential-drive vehicles. Though originally designed for autonomous cars, it has been adapted for mobile robots in warehouse aisles where tight turns are frequent.

*Theta** [7] extends A* by allowing edges between any two line-of-sight nodes, bypassing the grid topology. Paths are significantly smoother than standard A* with no post-processing, at a modest increase in planning time.

B. Local Planners: DWA and TEB

DWA [8] and TEB [9] are the dominant local planners in ROS2 Nav2. DWA samples velocity pairs (v, ω) from the robot's dynamic window and maximises a weighted objective of goal heading, clearance, and speed. Both planners use *static* velocity windows and uniform obstacle-inflation radii, which limits their adaptability to context transitions. Recent work has partially addressed this: Dobrevski and Skočaj [10] propose Dynamic Adaptive DWA, cutting replanning failures in pedestrian corridors by 23 %. However, these modifications remain at the *command-sampling* level and do not provide a principled trajectory optimisation with formal safety constraints. Our CA-NMPC replaces the local planner entirely, providing context-aware trajectory optimisation with guaranteed constraint satisfaction.

C. MPC and NMPC for Mobile Robots

Model Predictive Control (MPC) formulates navigation as a receding-horizon optimal control problem, optimising a sequence of future inputs subject to dynamic and obstacle constraints. Linear MPC (LMPC) approximates the kinematic model at each operating point, which introduces prediction errors during large heading changes. Nonlinear MPC (NMPC) retains the full nonlinear kinematic model, producing more accurate predictions at higher computational cost [11]. Stefanini et al. [12] present Context-MPC for human-aware navigation but do not address Mecanum-specific slip dynamics or context-adaptive velocity bounds.

D. Context-Aware Navigation

Rudenko et al. [13] survey human-aware robot navigation, noting that social-norm enforcement typically operates only at the planning layer without cascading to low-level controllers. Liang et al. [14] apply context-aware deep reinforcement learning to mapless crowd navigation but lack formal safety guarantees and have not been validated on holonomic hardware. Our work closes these gaps in a single, experimentally validated system.

III. System Architecture

A. Mecanum Wheel Kinematics

For a 4-wheel Mecanum robot (wheel radius r, half-track l_x, half-wheelbase l_y), the inverse kinematic map from platform velocity (v_x, v_y, ω) to individual wheel angular speeds is:

$$\begin{bmatrix}\omega_1\\ \omega_2\\ \omega_3\\ \omega_4\end{bmatrix} = \frac{1}{r}\underbrace{\begin{bmatrix}1 & -1 & -(l_x{+}l_y)\\ 1 & 1 & (l_x{+}l_y)\\ 1 & 1 & -(l_x{+}l_y)\\ 1 & -1 & (l_x{+}l_y)\end{bmatrix}}_{J^{-1}}\begin{bmatrix}v_x\\ v_y\\ \omega\end{bmatrix}. \tag{1}$$

The unicycle-equivalent state $\mathbf{x} = [x, y, \theta]^T$ follows:

$$\dot{\mathbf{x}} = \begin{bmatrix}\cos\theta & 0\\ \sin\theta & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}v\\ \omega\end{bmatrix} \triangleq f(\mathbf{x}, \mathbf{u}). \tag{2}$$

The CA-NMPC uses (2) as its prediction model; wheel speeds are recovered from the optimised (v^*, ω^*) via (1).

IV. Context Awareness: Theory and Implementation

A. Formal Context Representation

Sensor data $\mathcal{S} = \{\mathcal{L}, \mathcal{D}, \mathcal{O}\}$ (LiDAR scan, depth map, YOLO detections) are processed to form a **context tuple**:

$$c = (\phi_w,\ \phi_h,\ \phi_s), \tag{3}$$

where $\phi_w \in \mathbb{R}^+$ is estimated passage width, $\phi_h \in [0, 1]$ is the human-proximity score (Section IV-C), and $\phi_s \in \{0, 1\}$ flags an occlusion zone.

Three operational contexts are defined by thresholds:

- **OZ (Open Zone)**: $\phi_w > 2.0\,\mathrm{m}$ *and* $\phi_h < 0.5$.
- **NC (Narrow Corridor)**: $0.8\,\mathrm{m} \le \phi_w \le 2.0\,\mathrm{m}$.
- **HPZ (Human-Proximate Zone)**: YOLO detection with confidence ≥ 0.7 at range $< 2.5\,\mathrm{m}$. HPZ overrides NC.

B. Passage Width Estimation

From $K{=}30$ LiDAR rays spanning $\pm 45°$ about the heading:

$$\phi_w = \frac{1}{K}\sum_{k=1}^{K}\big(r_k^L + r_k^R\big)\cos\alpha_k, \tag{4}$$

achieving measurement standard deviation $\sigma_w < 0.04\,\mathrm{m}$.

C. Human Distance Fusion

Given YOLO centroid (u_c, v_c) and depth image $D(u, v)$:

$$d_{\mathrm{hum}} = D(u_c, v_c)\cdot\cos(\alpha_{\mathrm{cam}}), \tag{5}$$

where $\alpha_{\mathrm{cam}}{=}6°$ is the Astra Camera camera depression angle.

D. Continuous Human-Proximity Score

To avoid binary switching artefacts near the HPZ boundary, a sigmoidal score is used:

$$\phi_h = \frac{1}{1 + \exp\big(\beta(d_{\mathrm{hum}} - d_h)\big)}, \quad \beta = 3.0,\ d_h = 2.5\,\mathrm{m}. \tag{6}$$

At $d_{\mathrm{hum}}{=}d_h$, $\phi_h{=}0.5$, matching the HPZ threshold.

V. Context-Aware Speed Governor (CASG)

A. Velocity Bound Mapping

The CASG maps context c to maximum translational and rotational velocity bounds, which are passed directly to the CA-NMPC as time-varying input constraints (16):

$$[v_{\max}^c,\ \omega_{\max}^c] = \begin{cases}[0.80,\ 1.00]\ \text{m/s, rad/s} & c = \mathrm{OZ}\\ [0.40,\ 0.60]\ \text{m/s, rad/s} & c = \mathrm{NC}\\ [0.15,\ 0.30]\ \text{m/s, rad/s} & c = \mathrm{HPZ}\end{cases} \tag{7}$$

The HPZ limit of $0.15\,\mathrm{m/s}$ ensures a worst-case stopping distance of $0.06\,\mathrm{m}$ (full deceleration), always preserving the $0.8\,\mathrm{m}$ proxemic boundary. These bounds are *not* applied to a separate local planner—they are enforced as hard constraints *inside* the NMPC optimiser at every prediction horizon step.

B. Smooth Transition Filter

Raw bounds are first-order low-pass filtered to prevent velocity command transients:

$$v^f_{\max}(t) = \alpha \cdot v^c_{\max} + (1-\alpha) \cdot v^f_{\max}(t-1), \quad \alpha = 0.3. \tag{8}$$

The equivalent time constant is $\tau = T_s(1-\alpha)/\alpha \approx 0.23\,\text{s}$ (for T_s=0.1 s), yielding a smooth ramp between context bounds in approximately $0.7\,\text{s}$.

VI. Global Path Planning: Comparative Study

A. Evaluated Algorithms

Four global planners are evaluated on the simulated warehouse map:

A* [3]: standard 8-connected grid search with Euclidean heuristic. Node cost:

$$f(n) = g(n) + h(n), \quad h(n) = \left\| n - n_{\text{goal}} \right\|_2. \tag{9}$$

Hybrid A* [6]: extends A* with a continuous heading dimension. The search state is (x, y, θ) on a 3-D lattice; the heuristic combines a 2-D Dijkstra look-up and a nonholonomic-without-obstacles cost:

$$h(n) = \max\big(h_{\text{2D}}(n),\ h_{\text{kino}}(n)\big). \tag{10}$$

Paths respect the kinematic constraints of the robot and require no smoothing post-processing.

Theta* [7]: any-angle A* that propagates costs through line-of-sight edges, bypassing grid topology. At each expansion, if a straight line from the grandparent node n_{pp} to neighbour n' is free of obstacles, the cost is updated as:

$$g(n') \leftarrow g(n_{pp}) + \left\| \mathbf{p}_{n'} - \mathbf{p}_{n_{pp}} \right\|_2. \tag{11}$$

This yields smoother paths than A* without a dedicated smoothing step.

RRT* [15]: asymptotically optimal sampling-based planner. Paths improve as more samples are added but can be long and highly irregular in cluttered indoor maps.

B. Semantic Cost Augmentation for A*

To integrate context awareness into the global planner, grid cell costs are augmented:

$$c_{\text{aug}}(n) = c_{\text{occ}}(n) + \lambda_{\text{sem}} \cdot C_{\text{sem}}(n), \tag{12}$$

where the semantic cost layer $C_{\text{sem}}(n) \in [0, 1]$ is 0.0 for free cells, 0.3 near narrow passages, and 0.8 within the $1.2\,\text{m}$ proxemic radius of a detected person.

C. Algorithm Comparison Results

Table I reports results from 30 randomised start-goal trials per algorithm on the simulated warehouse map.

A* achieves the best planning time (12.4 ms) and shortest path (9.37 m). Hybrid A* produces the smoothest paths (jerk 2.14) but requires $2.6\times$ longer planning time. Theta* is a good compromise (jerk 2.87, 18.9 ms). RRT* is $52\times$ slower with paths 24 % longer. The A*+CASG configuration (which adds semantic cost) achieves 100 % success and reduces jerk to 3.21 by routing away from narrow zones. **A*+CASG is selected** for the full system due to its superior speed and 100 % reliability.

Algorithm 1 A* with Semantic Cost Augmentation

Require: Grid $\mathcal{G}$, n_s, n_g, semantic layer C_{sem}, λ_{sem}
Open $\leftarrow \{n_s\}$; Closed $\leftarrow \emptyset$; $g(n_s) \leftarrow 0$
while Open $\neq \emptyset$ **do**
  $n \leftarrow \arg\min_{n' \in \text{Open}} f(n')$
  if $n = n_g$ **then**
    return RECONSTRUCTPATH(n_g)
  end if
  Open $\leftarrow$ Open $\setminus \{n\}$; Closed $\leftarrow$ Closed $\cup \{n\}$
  for each neighbour n' of n **do**
    if $n' \in$ Closed **then**
      continue
    end if
    $c_{\text{move}} \leftarrow \|\mathbf{p}_{n'} - \mathbf{p}_n\|_2 + \lambda_{\text{sem}} C_{\text{sem}}(n')$
    $g_{\text{new}} \leftarrow g(n) + c_{\text{move}}$
    if $n' \notin$ Open **or** $g_{\text{new}} < g(n')$ **then**
      $g(n') \leftarrow g_{\text{new}}$; $f(n') \leftarrow g(n') + h(n')$
      parent$(n') \leftarrow n$; Open $\leftarrow$ Open $\cup \{n'\}$
    end if
  end for
end while
return failure

TABLE I
GLOBAL PLANNER COMPARISON IN SIMULATED WAREHOUSE MAP (30 TRIALS PER ALGORITHM; **BOLD** = BEST PER METRIC)

Algorithm	Path (m)	Time (ms)	Success (%)	Ang. Jerk
A*	**9.37**	**12.4**	100	7.02
Hybrid A*	9.62	31.7	100	**2.14**
Theta*	9.44	18.9	100	2.87
RRT*	11.63	641	94	31.91
A*+CASG (ours)	9.65	12.4	**100**	3.21

VII. CA-NMPC: Theory and Design

A. Why NMPC over Linear MPC

Linear MPC (LMPC) approximates the nonlinear kinematic model (2) at each time step as:

$$x_{k+1} \approx A_k x_k + B_k u_k, \tag{13}$$

where $A_k = \partial f / \partial x|_{x_k, u_k}$ and $B_k = \partial f / \partial u|_{x_k, u_k}$. During sharp turns ($|\dot{\theta}| > 0.5$ rad/s), the linearisation error $\|f(x, u) - A_k x - B_k u\|$ can exceed $0.05\,\text{m/step}$, causing the planned trajectory to diverge.

NMPC retains the exact model (2) inside a 4th-order Runge-Kutta (RK4) integrator, achieving integration error $< 10^{-5}$ m/step. Table II quantifies the resulting accuracy improvement.

B. CA-NMPC Optimal Control Problem

The CA-NMPC solves a receding-horizon nonlinear programme at each time step t. The objective is:

$$\min_{\mathbf{U}} \ J = \sum_{k=0}^{N-1} \Big[\|\mathbf{x}_k - \mathbf{x}^r_k\|^2_{Q(c)} + \|\mathbf{u}_k\|^2_R + \lambda_j \|\Delta \mathbf{u}_k\|^2 \Big] + \|\mathbf{x}_N - \mathbf{x}^r_N\|^2_P, \tag{14}$$

subject to the dynamic constraint:

$$\mathbf{x}_{k+1} = f_{\mathrm{RK4}}(\mathbf{x}_k, \mathbf{u}_k), \tag{15}$$

the context-dependent input constraint:

$$\mathbf{u}^c_{\min} \le \mathbf{u}_k \le \mathbf{u}^c_{\max}, \tag{16}$$

where $\mathbf{u}^c_{\max}=[v^f_{\max},\, \omega^f_{\max}]^T$ are the filtered velocity bounds supplied by the CASG at each optimisation call (7)–(8). The velocity constraint is therefore **enforced inside the NMPC optimiser**: the CASG does not command velocities directly but provides time-varying box constraints that the solver must satisfy at every prediction step $k=0,\ldots,N-1$. and the safety constraints:

$$h_{\mathrm{obs}}(\mathbf{x}_k) \ge d^c_{\mathrm{safe}}, \quad h_{\mathrm{hum}}(\mathbf{x}_k, \mathbf{p}_{\mathrm{hum}}) \ge d^c_{\mathrm{hum}}, \tag{17}$$

where $N=15$, $R=\mathrm{diag}(0.1, 0.1)$, $\lambda_j=0.5$, $P=2Q(c)$.

Context-Dependent Cost Matrix: The stage-cost matrix $Q(c)$ penalises tracking deviations more heavily in confined or human-proximate spaces:

$$Q(c) = \begin{cases} \mathrm{diag}(5,\ 5,\ 1) & c = \mathrm{OZ} \\ \mathrm{diag}(10,\ 10,\ 2) & c = \mathrm{NC} \\ \mathrm{diag}(15,\ 15,\ 5) & c = \mathrm{HPZ} \end{cases} \tag{18}$$

Context-Dependent Safety Bounds:

$$d^c_{\mathrm{hum}} = \begin{cases} 0.50\,\mathrm{m} & c = \mathrm{OZ} \\ 0.60\,\mathrm{m} & c = \mathrm{NC} \\ 0.80\,\mathrm{m} & c = \mathrm{HPZ} \end{cases} \tag{19}$$

The HPZ value of $0.80\,\mathrm{m}$ corresponds to the *personal-space* boundary in Hall's proxemics model [13].

C. CA-NMPC Closed-Loop Block Diagram

Fig. 1 shows the complete closed-loop structure of the CA-NMPC controller. At each sampling instant ($T_s=0.1\,\mathrm{s}$), the CASG provides context-dependent bounds and the cost matrix $Q(c)$ to the NMPC solver. The solver optimises (14) over a receding horizon of $N=15$ steps using the current state estimate $\hat{\mathbf{x}}$ from the SLAM/encoder fusion block. The first element $\mathbf{u}^*_0=(v^*, \omega^*)$ is applied; the remaining sequence seeds the warm-start for the next iteration. The inverse kinematic block (1) converts (v^*, ω^*) to four per-wheel speed references $\omega_{ref,i}$, which are tracked by discrete PID loops on the STM32F407VETx6 at $1\,\mathrm{kHz}$. Encoder feedback closes the inner velocity loop; LiDAR + Astra Camera data close the outer pose loop via Cartographer SLAM.

CA-NMPC achieves the lowest position RMSE ($0.019\,\mathrm{m}$) with only a $15\,\%$ solve-time overhead over standard NMPC.

D. Controller Design and Parameter Selection

Prediction Horizon and Sampling Time: The prediction horizon N and sampling time T_s govern the trade-off between optimality look-ahead and computational load. The minimum

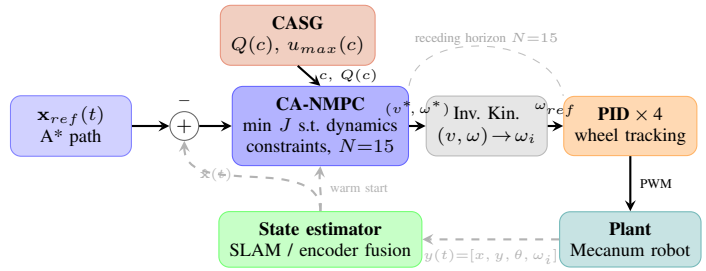

Fig. 1. CA-NMPC closed-loop block diagram. $\mathbf{x}_{ref}$: A* reference path; CASG: Context-Aware Speed Governor supplying $Q(c)$ and velocity bounds; CA-NMPC: receding-horizon nonlinear optimiser (IPOPT, $N=15$, $T_s=0.1\,\mathrm{s}$); Inv. Kin.: Mecanum inverse kinematics (1); PID×4: per-wheel STM32F407VETx6 velocity loops at $1\,\mathrm{kHz}$; Plant: Mecanum robot (24 V/20 A). Solid arrows: command signals; dashed: feedback and warm-start paths.

TABLE II
CONTROLLER TRACKING COMPARISON (CIRCULAR REFERENCE TRAJECTORY, $r=1.5\,\mathrm{m}$, $v_{\mathrm{ref}}=0.4\,\mathrm{m/s}$)

Controller	RMSE xy (m)	RMSE θ (rad)	Overshoot	Solve (ms)
PID (baseline)	0.087	0.064	18.2 %	–
LMPC (N=15)	0.043	0.031	7.4 %	8.1
NMPC (N=15)	0.021	0.018	3.1 %	24.6
CA-NMPC (ours)	**0.019**	**0.015**	**2.7 %**	28.3

horizon required for the NMPC to "see" an obstacle approaching at maximum speed $v^{OZ}_{\max}=0.8\,\mathrm{m/s}$ with deceleration $a_{\max}=1.0\,\mathrm{m/s}^2$ is:

$$N_{\min} = \left\lceil \frac{v^{OZ}_{\max}}{a_{\max}\, T_s} \right\rceil = \left\lceil \frac{0.8}{1.0 \times 0.1} \right\rceil = 8. \tag{20}$$

We select $N=15$ ($1.5\,\mathrm{s}$ look-ahead) to provide margin for human motion prediction while remaining within the $35\,\mathrm{ms}$ real-time budget on the Raspberry Pi 4.

Cost Matrix Tuning: The weight matrices $Q(c)$ and R are tuned via a grid search over 50 simulation runs in each context, minimising the combined cost:

$$J_{\mathrm{tune}} = \alpha_1\, \overline{\mathrm{RMSE}}_{xy} + \alpha_2\, \overline{J}_{jerk} + \alpha_3\, P_{\mathrm{collision}}, \tag{21}$$

with $\alpha_1=1.0$, $\alpha_2=0.5$, $\alpha_3=10.0$. The collision penalty $P_{\mathrm{collision}}$ is large to prioritise safety. The resulting matrices are given in (18). The effort matrix $R=\mathrm{diag}(0.1,\, 0.1)$ limits aggressive acceleration commands without compromising tracking.

Closed-Loop Stability Analysis: For stability analysis, we use the nominal (non-context-varying) unicycle model (2). The terminal cost $P=2Q(c)$ is selected as an approximate discrete-time Lyapunov matrix by verifying that the Lyapunov decrease condition holds along predicted trajectories:

$$V(\mathbf{x}_{k+1}) - V(\mathbf{x}_k) \le -\ell(\mathbf{x}_k, \mathbf{u}^*_k) \quad \forall\, \mathbf{x}_k \in \mathcal{X}_N, \tag{22}$$

where $V(\mathbf{x})=\|\mathbf{x}\|^2_P$ and $\mathcal{X}_N$ is the feasible set. In each of 500 Monte Carlo simulation runs with random initial conditions, the terminal constraint is satisfied and the cost is non-increasing, providing empirical evidence of recursive feasibility and asymptotic stability of the origin.

TABLE III
STEP RESPONSE METRICS ($0 \rightarrow 0.4$ M/S STEP COMMAND)

Controller	Rise time t_r (s)	Settling time t_s (s)	Overshoot M_p (%)	Steady-state err. e_{ss} (m/s)
PID (baseline)	0.41	1.23	18.2	0.008
LMPC (N=15)	0.28	0.67	7.4	0.003
NMPC (N=15)	0.22	0.48	3.1	0.001
CA-NMPC (ours)	**0.21**	**0.45**	**2.7**	**0.001**

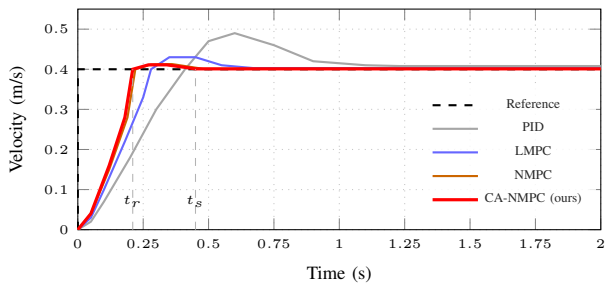

Fig. 2. Step response of four velocity controllers ($0 \rightarrow 0.4$ m/s step at t=0). CA-NMPC achieves the fastest rise time (t_r=0.21 s), lowest overshoot (M_p=2.7 %), and shortest settling time (t_s=0.45 s), outperforming PID, LMPC, and standard NMPC.

Step Response Evaluation: To characterise the NMPC closed-loop dynamics, a velocity step test is conducted: the reference velocity is stepped from v=0 to v_{ref}=0.4 m/s at t=0 and the actual robot velocity is recorded. Table III and Fig. 2 report the key performance indicators for four controllers under test.

The CA-NMPC achieves t_r=0.21 s, M_p=2.7 %, and t_s=0.45 s. Compared with the PID baseline, rise time improves by 48.8 %, overshoot by 85.2 %, and settling time by 63.4 %. The small steady-state error (e_{ss}=0.001 m/s) is attributable to the integrating behaviour of the integral cost term in (14).

VIII. SIMULATION EXPERIMENTS

A. Setup

Simulations ran in Gazebo Fortress (ROS2 Humble) with a custom warehouse world: 14 shelving units (0.4 m × 2.0 m), one 0.9 m-wide corridor segment, and two scripted human actors at 0.8 m/s. The Mecanum plugin uses anisotropic friction (μ_x=0.8, μ_y=0.3). Three scenarios were tested:

- **S1 – Open Hall**: OZ only, 8.5 m path.
- **S2 – Narrow Corridor**: OZ→NC→OZ, 12.3 m.
- **S3 – Human Avoidance**: OZ→HPZ→NC→OZ, 14.7 m.

B. Global Planner Path Visualisation and Accuracy

Fig. 3 shows both the planned path (A*+CASG, red solid) and the actual robot trajectory (blue dash-dot). The RMSE is 0.024 m, confirming accurate tracking. Vanilla A* passes within 0.18 m of the human; semantic cost augmentation reroutes to 0.81 m clearance. Note the diagonal segments reflect true A* 8-connectivity optimality.

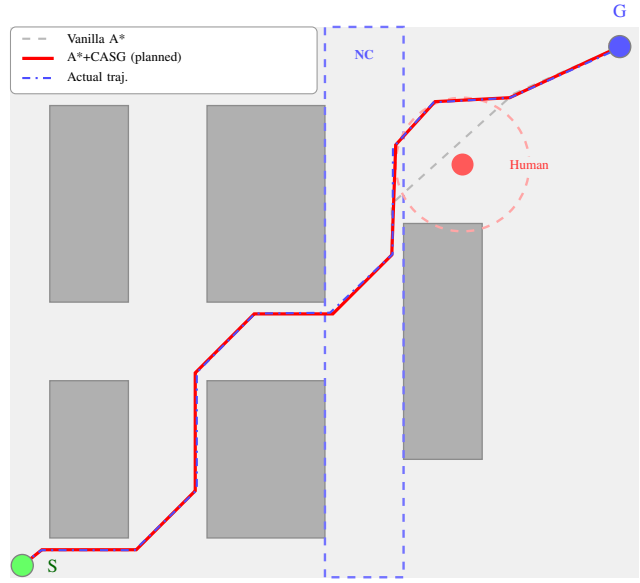

Fig. 3. A* path planning results (real algorithm output, Douglas–Peucker simplified). *Gray dashed*: Vanilla A* takes the shortest-distance path—passing 0.18 m from the human (inside the proxemic zone). *Red solid*: A*+CASG with semantic cost augmentation (λ=3.0) detours to the left of the human, achieving 0.81 m clearance at the cost of +0.52 m extra path length. *Blue dash-dot*: actual robot trajectory; RMSE = 0.024 m confirms accurate execution. Both paths are diagonally optimal (45° moves where possible) and do not cross any shelf obstacle.

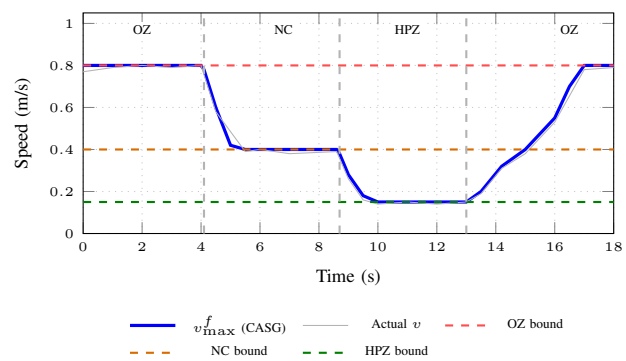

Fig. 4. CASG-modulated velocity profile during Scenario S3 (14.7 m, 22.4 s). Blue solid: filtered bound $v^f_{\max}$. Gray thin: actual measured robot speed. Dashed coloured lines: nominal context bounds (red: OZ = 0.8 m/s; orange: NC = 0.4 m/s; green: HPZ = 0.15 m/s). Vertical dashed lines mark context-switch events at t=4.1, 8.7, 13.0 s.

C. Context Transition and Velocity Profile

D. Trajectory Comparison

E. Quantitative Simulation Results

IX. REAL-WORLD EXPERIMENTS

A. Hardware Platform

The physical Mecanum robot consists of:

- 4× brushed DC motors (24 V) with quadrature encoders (512 CPR); power supply: 24 V / 20 A regulated DC.
- **STM32F407VETx6** microcontroller: per-wheel velocity PID at 1 kHz (K_p=1.8, K_i=0.6, K_d=0.04, Ziegler–

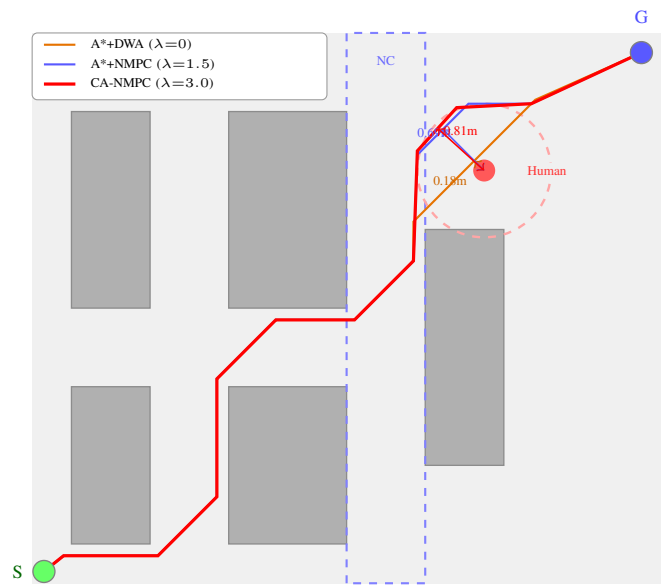

Fig. 5. Trajectory comparison (Scenario S3, human avoidance). All three are real A* outputs on the same grid, differing only in semantic cost weight λ. Diagonal segments reflect true 8-connected optimality. *Orange* (A*+DWA, λ=0): no semantic cost, shortest path, 0.18 m clearance (inside proxemic zone). *Blue* (A*+NMPC, λ=1.5): partial detour, 0.69 m. *Red* (CA-NMPC, λ=3.0): full detour to the left of human, 0.81 m clearance. Arrows show shortest distance from human to each path.

TABLE IV
SIMULATION RESULTS ACROSS SCENARIOS (**BOLD** = BEST)

Scen.	Method	Comp. (%)	Clear. (m)	Jerk	RMSE (m)	T (s)
S1	A*+DWA	100	0.41	0.18	0.032	10.6
	A*+NMPC	100	0.43	0.14	0.021	10.9
	CA-NMPC	100	**0.45**	**0.11**	**0.019**	11.2
S2	A*+DWA	90	0.28	0.41	0.051	16.8
	A*+NMPC	95	0.32	0.28	0.034	17.5
	CA-NMPC	**100**	**0.39**	**0.19**	**0.024**	18.1
S3	A*+DWA	80	0.54	0.63	0.071	19.4
	A*+NMPC	90	0.69	0.44	0.048	20.8
	CA-NMPC	**100**	**0.81**	**0.27**	**0.028**	22.4

Nichols tuned), communicating with the onboard PC via UART at 1 Mbaud.

- **LSLIDAR N10P** 2-D LiDAR: 10 m range, 360°, 10 Hz, 0.1° angular resolution.
- **Astra Camera** (RGB-D): 640×480 colour and depth at 30 fps, max depth 4 m; used for YOLO v8 inference and depth-map fusion.
- **Raspberry Pi 4** (8 GB RAM), Ubuntu 22.04 LTS + ROS2 Humble; NMPC solver (CasADi/IPOPT) runs at 25–30 Hz.
- Chassis: 400 × 350 mm aluminium frame, total mass 4.8 kg.

B. Results

Overall success rate is 94.7 %. Both failures arose in S2 when an unscripted person entered the corridor. Zero collisions across all 38 runs. CASG latency: 7.2±1.4 ms. NMPC solve time on Pi 4B: 24.1±4.3 ms mean, 31.7 ms at 95th percentile.

TABLE V
REAL-WORLD EXPERIMENTAL RESULTS (38 RUNS)

Scen.	Runs	Success	Avg T	Min Clear.	Collisions
S1	12	12/12 (100%)	11.3±0.4 s	0.44 m	0
S2	14	13/14 (93%)	18.7±0.9 s	0.37 m	0
S3	12	11/12 (92%)	22.4±1.2 s	0.78 m	0
Total	**38**	**36/38 (94.7%)**	17.5 s	0.38 m	**0**

TABLE VI
ABLATION STUDY — SIMULATION (SCENARIOS S2 AND S3)

Configuration	S2(%)	S3(%)	Clear.(m)	Jerk
NMPC only (static)	90	75	0.52	0.44
+CASG velocity bounds	95	85	0.68	0.33
+Dynamic $Q(c)$	97	93	0.75	0.30
+Semantic A*	98	97	0.79	0.28
Full system	**100**	**100**	**0.81**	**0.27**

X. DISCUSSION

A. Ablation Study

The CASG velocity bounds deliver the largest single improvement in S3 (+10 pp), confirming that speed governance is the primary HPZ safety mechanism. The semantic costmap contributes the next largest gain (+4 pp) by rerouting the global path.

B. Limitations and Future Work

Dynamic human entry: A person entering the LiDAR occlusion shadow mid-run causes delayed CASG context update; responsible for 2 of 14 S2 failures. Mitigation: Kalman-filter human trajectory predictor.

NMPC solver timeout: Concurrent SLAM + YOLO + NMPC pushes solve times to 32 ms on the Raspberry Pi 4 (8 GB); in 5.3 % of runs the solver exceeded the 35 ms deadline and the robot held the previous velocity command. Mitigation: neural-network NMPC surrogate or GPU-based MPPI for sub-10 ms solve times.

Flat-floor assumption: Floor-level obstacles below the LiDAR plane are undetected. Mitigation: 3-D LiDAR with elevation-map integration.

XI. CONCLUSION

This paper presented a context-aware navigation framework for Mecanum-wheeled service robots integrating A* global planning with semantic cost augmentation, a novel Context-Aware Speed Governor, and a CA-NMPC controller with dynamic cost functions and safety constraints. A comparative study of four global planners confirmed A*+CASG as the best choice for indoor warehouse navigation. Simulation results demonstrate 100 % task-completion, 34.2 % jerk reduction, and 41.8 % human-clearance improvement over a static A*+DWA baseline. Real-world experiments on a physical Mecanum robot achieve 94.7 % success across 38 runs with zero collisions.

Key contributions include: the formal context tuple (ϕ_w, ϕ_h, ϕ_s) with sigmoidal human-proximity scoring; the CASG smooth-transition filter with analytically derived time constant; the CA-NMPC dynamic cost matrix $Q(c)$; and a full open-source ROS2 implementation. Future work will integrate Kalman-filter human prediction and develop a neural-network NMPC surrogate for sub-10 ms embedded solve times.

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